Volatility Modelling and Risk Analysis of Gold Futures in R, 2022 – April 2026

Chapter 1: set-up, data scraping and transformation

The set-up

The ***R.code*** file presents the required packages, of which 'quantmod' proves to be an essential part of an Econometric analysis.

Data Scraping

Setting from = "YYYY/MM/DD" and to = "YYYY/MM/DD" for commodity GC=F scrapes Gold future prices for a time period based upon recent market data. Two sets are acquired, a four-year dataset and another with all trading days since 2002.

In this case, the four-year set is written as a CSV called 'data'. Users are encouraged to repeat using sys=Date for rolling data (up to the present).

Transformation

Log returns are taken for the data. This includes the four year and 24-year datasets. Such a **transformation** is calculated by taking the differenced log closing prices for Gold. It follows the following equation:

$$r_t = Ln(P_t) - Ln(P_{t-1}) \quad (1)$$

This transformation (log) is useful for its statistical tractability and returns are scale-free which makes comparison independent-of-currency. Comparing figures 1A and 2A, to 3A and 4A shows the merit in using log returns.

Stationarity test

The ***ADF*** (Augmented Dickey-Fuller) test checks for a unit root, helping determine whether a time series is ***Stationary***.

Stationary series have stable statistical properties (e.g. constant mean and variance), which makes ARIMA analysis appropriate because lag relationships remain stable over time.

Weakly stationary Process refers to when a time series **{Xt}** or **{GC=Ft}** has statistical properties which are as follow:

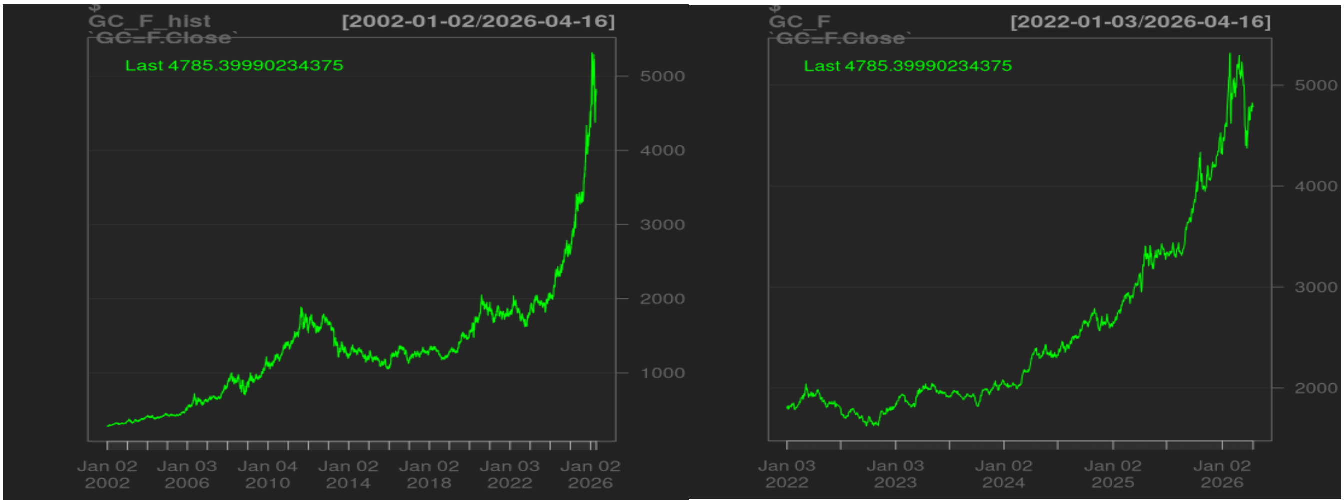
1) **Constant mean: E[Xt]** is independent of time **(t)**.

2) **Cov(Xt, Xt-k)** depends only on lag size '**k**'.

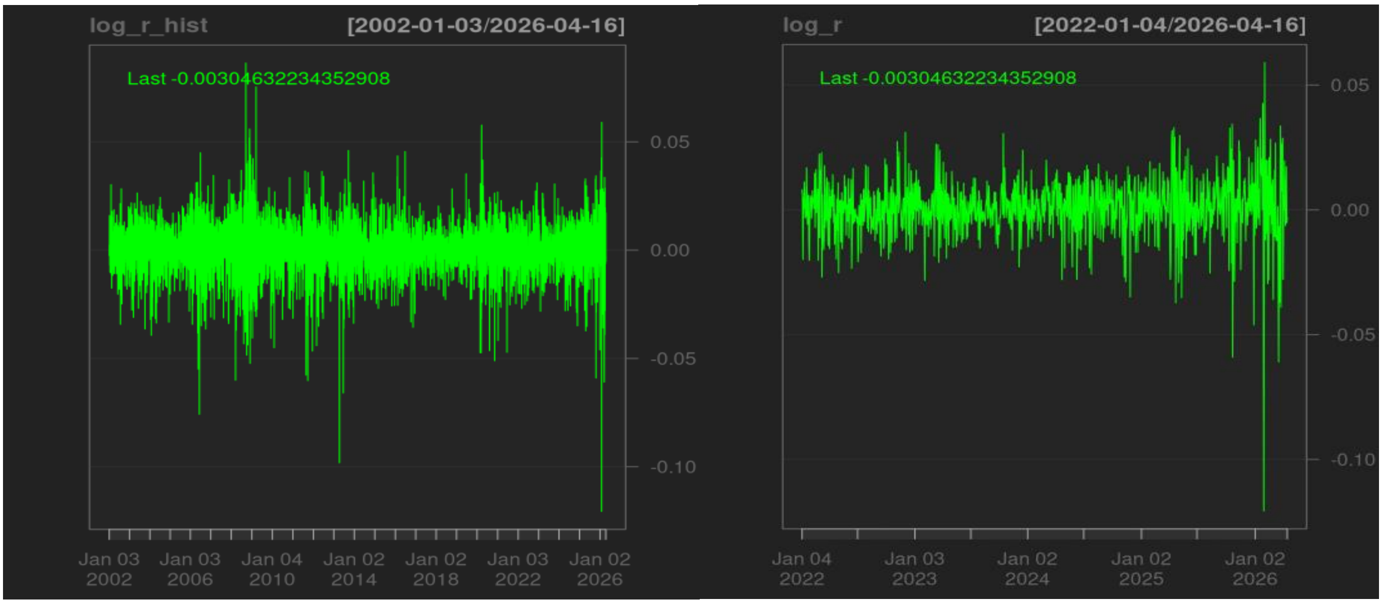
First ADF test on closing prices illustrates the merit of transformed metrics. Closing prices have a p-value of 0.9368. Resultingly, the null hypothesis of non-stationarity does not have sufficient grounds to be rejected. ARIMA modelling would be discouraged.

Second ADF test on log-returns was significant to 0.05 which provided sufficient evidence to reject Non-Stationarity at the 5% Confidence Interval. Reasonable grounds for Stationarity (alternative hypothesis). ARIMA modelling tools can be explored.

Figures 1A and 1B (24 year and four-year closing prices respectively)



Figures 1C and 1D (24 year and four-year log returns respectively)



Visualisations also support the ***ADF test results*** since ***1C*** and ***1D*** graphs appear to have a stable mean around zero. The build-up to January 2026 appears dramatic in ***Figure 1D*** but in the full time-series shown in ***Figure 1C***, similar turbulent periods occurred (2008 and 2021). Namely the 'Credit-crunch' and Covid crises.

Chapter 2: Distributive properties and VaR of log-returns

In the absence of trends, forecasting with ARIMA is possible (***chapter 3***). It may be tempting to skip but ***chapter 2*** provides useful descriptive statistics for understanding the data. This is a quick grasp of evolving market behaviour in a set-period (01/Jan/2022 to 17/Apr/2026).

Table 1, log returns parameters

Sample Parameter	Value	Rounded (2sig-fig)
Mean	0.000909	0.0010
Median	0.00109	0.0011
Variance (Standard Deviation)	0.000139, 0.0118	0.00014, 0.012
Skewness	-1.265	-1.3
Kurtosis - 3	9.31	9.30

VaR, Value at Risk (historic method)

For those wanting to understand more about VaR, please see (Investopedia):

Understanding Value at Risk (VaR): Explanation and Calculation Methods

Using the historical data, a level of certainty that losses will not exceed a given value can be determined. However, the ***VaR*** will not explain what occurs when losses exceed an unlikely threshold.

E.g. at 95% Confidence, what quantity losses reach is determined by a value represented within the worst 1/20 days. For 99% this is the worst 1/100 days.

Log returns (r_t) can be converted to Regular returns (R_t)

Reversing log returns for interpretability is advised. This is simple and shows how a following 'bad-day' can hurt a Portfolio or Investment by showing the minimum amount (%) that can be expected to be lost up to some level of confidence based on historical returns.

$$R_t = e^{(r_t)} - 1, \qquad (2)$$

ES, Expected Shortfall

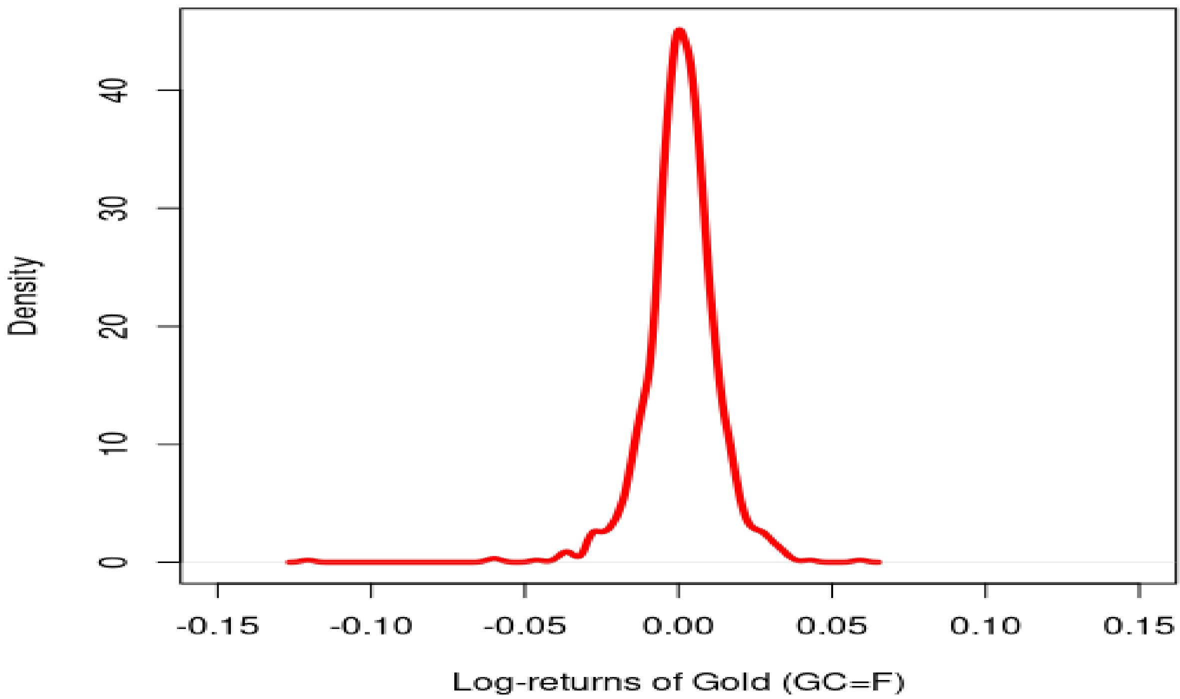
ES covers what is missing from ***VaR*** which is a loss-quantity including even the worst of trading days. For those interested, a link to a good resource for ***Expected Shortfall*** is below:

Expected Shortfall: Definition, Calculation, Use Cases & Key Advantages — marketopia.org

Table 2, VaR

Loss level (%)	***VaR***	***Conversion to returns, Rt (%)***
Max loss (0, quantile)	-0.1207, worst day in data **(Loss)**	-12.82%
One percent (0.01, quantile)	-0.029, Worst 1/100 days **(Loss)**	-2.90%
Five percent (0.05, quantile)	-0.0171, Worst 1/20 days **(Loss)**	-1.69%
Ten percent (0.1, quantile)	-0.0061, Worst 1/10 days **(Loss)**	-0.608%
Best day in data (1, quantile)	0.00591, Highest return **(Gain)**	6.1%

Figure 2A, VaR shown on density plot (log returns)



The density plot of Gold futures log-returns exhibits a pronounced central peak and relatively heavy tails, indicating a ***leptokurtic distribution***, which is coherent with the ***chapter 2 results***. This suggests that extreme price movements occur more frequently than would be expected under normality assumptions, a common characteristic of financial return series.

Chapter 3, Forecasting with AIC criterion

The appropriate ***ARIMA model*** can be constructed after investigating the Autocorrelation function ***(ACF)*** and Partial Autocorrelation function ***(PACF)***.

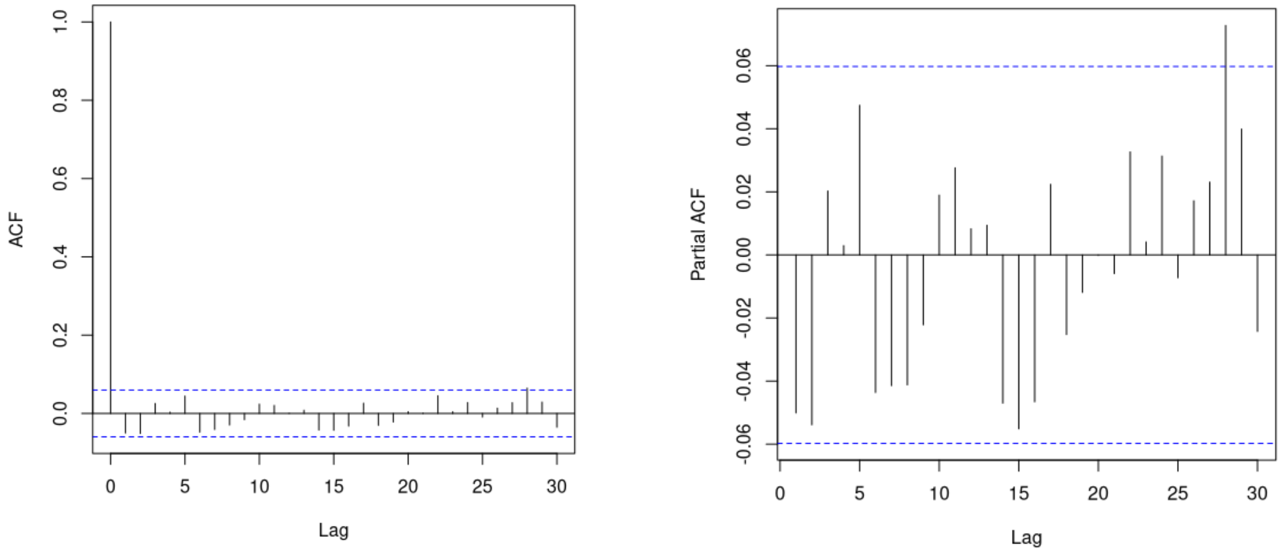
A rejection region is set automatically by R but it may be manually calculated as the following for the 95% Significance Level:

$$(1.96) / \sqrt{n} \qquad (3)$$

Where ***n*** is the number of observations. In this case, 0.0597 or approximately **6%**.

Then the ***ACF*** and ***PACF*** curves may be used to test against autocorrelations and see if there are significant lags.

Figures 3A and 3B, ACF and PACF respectively;



Autocorrelation function (3A) illustrates that the only relevant lag (lag 0 is always =1) was 28 which is most likely noise, unless some omitted 28-day seasonal effect is omitted. ***Partial ACF (3B)*** is coherent with only one lag significant which is 28. With 30 lags and 0.05 chance of error, then there may be ***0.05*30 = 1.5*** meaning one-to-two false positives ***(Type I errors)*** concluding that not much attention should be paid to lag-28. These results suggest that an ARIMA(0,0,0) should be most appropriate.

Nevertheless, three models are fitted and tested using Akaike's Information Criterion.

- Moving Average (p = 0, I = 0, q = 1) = ARIMA(0,0,1)
- Autoregressive (p = 1, I = 0, q = 0) = ARIMA(1,0,0)
- Mean-average (p = 0, I = 0, q = 0) = ARIMA(0,0,0)

Akaike's Information Criterion:

Balanced assessment of the fit versus complexity trade-off. Overly complex models risk overfitting and poor generalisability. An explanation below.

$$AIC = 2k - 2\ln(L) \qquad (4)$$

K represents the number of parameters, ***-L*** is the likelihood where lower is indicative of better-fit.

Low ***AIC scores*** are a sign of good fit for the level of complexity, near-values are equivalent. Explained by the larger spread between ***-2ln(L)*** and ***2k.***

Table 3, AIC ARIMA scores

Model	***AIC score***
Autoregressive, ARIMA (1,0,0)	-6461.70
Moving-Average, ARIMA (0, 0, 1)	-6492.02
White Noise Process, ARIMA (0,0,0)	-6491.01

Despite the results suggesting that Gold log returns follow a ***White Noise Process***, the best model fit under ***AIC*** was **ARI*MA(0,0,1)*** followed by **ARIMA(1,0,0)**. However, the fit for ARIMA(0,0,1) and ARIMA(1,0,0) are only marginally better, and given the principle of parsimony, the simpler ARIMA(0,0,0) should be preferred.

These results motivate the next section addressing ***Volatility***. Despite the log return values being almost entirely unpredictable based on prior lags, this does not interpolate to ***Volatility*** which may still be closely associated with its previous values.

Chapter 4, Volatility modelling

ARCH(1), introduced by Robert F. Engle in 1982, was developed to model time-varying ***Volatility*** in financial time series where the assumption of homoskedasticity is violated. In such processes, the conditional variance is not constant over time and periods of high ***Volatility*** tend to cluster together. Clustering raises the question of whether market shocks dissipate rapidly or persist over extended periods. Extending Engle's framework, Tim Bollerslev proposed the ***GARCH(1,1)*** model in 1986, allowing current ***Volatility*** to depend on both past squared innovations and previous conditional variances

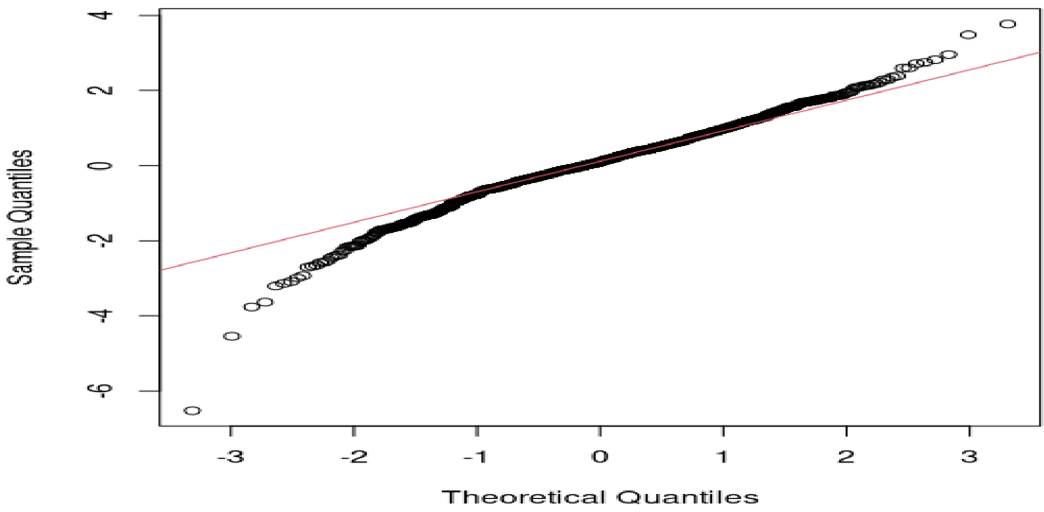
Upon fitting the ***GARCH(1,1)*** model, the volatility coefficients ***α0***, ***α1*** and ***β1*** are shown to all be significant to the 0.1% Significance level with respective coefficients at 0.00000323, 0.08885 and 0.8905.

- α_0 illustrates the level of ***Volatility*** that is constant which is small relative to dynamic volatility of ***ARCH*** α_1 and ***GARCH*** β_1 which combined are almost 0.97 or 1 illustrating that ***Volatility*** is highly persistent and effects decay slowly.
- α_1 covers the impact of ***new market innovations/information***. Such as the news which has long been understood as a component of ***Volatility*** (Mindell, 1961).
- β_1 shows how ***Volatility*** is subdued and ***depends highly on previous periods***.

$$\sigma_t\text{^}2 = \alpha_0 + \alpha(\varepsilon_{t-1})\text{^}2 + \beta(\sigma_{t-1})\text{^}2 \tag{5}$$

The ***ARCH*** term α_1 and ***GARCH*** term β_1 combine to almost 1. This result, along with all three terms individually is of high interest as it demonstrates the behaviour of market turbulence. These outcomes should be compared to out of sample periods to investigate how robust these results are during different periods in the Economic cycle.

Figure 4A, Normal QQ-plot of GARCH(1,1) residuals



The Normal Q-Q plot indicates substantial departures from Normality in the tails of the distribution. While the central quantiles align reasonably closely with the theoretical normal line, the extreme quantiles diverge significantly, illustrating the ***excess kurtosis*** in ***chapter 2*** of **9.31**. Such characteristics imply a greater probability of extreme market movements than predicted by the normal distribution.

Chapter 5: Suggestions/Limits

A suggestion is to model with the ***Student-t distribution***. Which is adapted for analysis on heavier tailed distributions. This was apparent from the ***excess Kurtosis*** results in **chapter 2**. At 9.31, this was noticeably different from zero.

Another suggestion is to account for the asymmetries present in the ***Skewness*** of ***chapter 2*** at -1.265 which demonstrated that the distribution was not symmetrical. Otherwise ***Skewness*** is equal to zero, and since this was not the case, the ***GARCH(1,1)*** model failed to account for the 'leverage effect', when negative news has a larger impact than positive news/innovations (Braun et al,. 1995). This would mean that adapted versions of the ***GARCH*** family such as ***EGARCH*** (Nelson, 1991) and ***GJR-GARCH*** (Glosten, 1993) are suggested for addressing the asymmetries of the log-returns.

Historical Value at Risk (VaR) would capture some natural variability but may be prone to overfitting on new data. ***Overfitting*** can hinder robustness, especially if out-of-sample results show that ***VaR*** is very different in other historical periods. Additionally,

VaR does not account for losses beyond the Confidence Interval. For this reason, ***Expected Shortfall (ES)*** would be a valuable tool.

Links and Citations

Value at Risk, Investopedia website (25, November, 2025):
https://www.investopedia.com/articles/04/092904.asp

Expected Shortfall, Marketopedia website, (21, April, 2026):

https://www.marketopia.org/blog/expected-shortfall/

Bollerslev, T., 1986. *Generalized autoregressive conditional heteroskedasticity. Journal of econometrics, 31(3), pp.307-327.*

Braun, P.A., Nelson, D.B. and Sunier, A.M., 1995. *Good news, bad news, volatility, and betas. The Journal of Finance, 50(5), pp.1575-1603.*

Engle, R.F., 1982. *Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. Econometrica: Journal of the econometric society, pp.987-1007.*

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Mindell, J., 1961. *How news affects market trends. Financial Analysts Journal, 17(1), pp.31-34.*

Nelson, D.B., 1991. *Conditional heteroskedasticity in asset returns: A new approach. Econometrica: Journal of the econometric society, pp.347-370.*